9. Two Stage Least Squares and Modern IV Estimators

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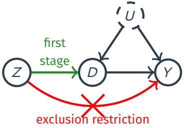
Outline

Where are we? Where are we going?

- Last time:
    - **Instrumental variable** under noncompliance in randomized experiments.
        - The local ATE (or CACE): $\tau_{\text{LATE}} = \text{ITT}_{Y,\text{co}} = \frac{\text{ITT}_Y}{\text{ITT}_D}$
        - The Wald/IV estimator: $\widehat{\tau}_{\text{iv}} = \widehat{\text{ITT}}_Y / \widehat{\text{ITT}}_D$
        - Intent-to-treat analysis, compliance types, identification assumptions..



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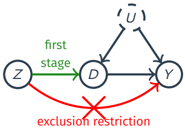

- Today:
  1. **Two-Stage Least Squares** (TSLS): estimands, weak IVs, multivalued D, general 2SLS.
  2. **Modern IV via Machine Learning**: PLIV (constant τ), DRIV (heterogeneous $\tau(\mathbf{X})$), DML.
  3. **Estimating IV in practice**: AJR walkthrough + modern IV in R.
  4. **Notable IV designs**: treatment assignments, regional characteristics, peers' environments, shift-share.

Source: *Chapter* 4 of Mostly Harmless Econometrics (Textbook 1) by J. Angrist & J. Pischke

1/ Two-Stage Least Squares

Two Stage Least Squares

- **Two-stage least squares** (TSLS) is the classical approach to IV which assumes two linear models:

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- But we have an **instrument** Z_i that is exogenous $\mathbb{E}[\varepsilon_i | Z_i] = 0$.
  - It's also exogenous for treatment uptake, so $\mathbb{E}[\eta_i | Z_i] = 0$.
- This implies the following CEF form for Y_i conditional on Z_i:

$$\mathbb{E}[Y_i | Z_i] = \alpha + \tau \mathbb{E}[D_i | Z_i] = \alpha + \tau \cdot (\gamma Z_i)$$

  - α is reused loosely (technically $\alpha + \tau\delta$ here); not the parameter of our interest.

TSLS Estimands

- Under the model, we have the following CEF: $\mathbb{E}[Y_i | Z_i] = \alpha + \tau \cdot (\gamma Z_i)$
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- TSLS estimator:
  - Estimate $\widehat{\gamma}$ from regression of treatment D_i on instrument Z_i
  - Estimate $\widehat{\tau}_{2SLS}$ as the slope of a regression of Y_i on $\widehat{\gamma} Z_i$.
    - $\widehat{D}_i = \widehat{\delta} + \widehat{\gamma} Z_i$; the Z-driven $\widehat{\gamma} Z_i$ is exogenous, uncorrelated with ε_i (constant $\widehat{\delta}$ doesn't affect the slope).
  - Under this model, $\widehat{\tau}_{2SLS} \xrightarrow{p} \tau$ (but don't use SEs from second stage; see MHE section 4.6.1. *2SLS Mistakes*)

Binary Treatment and Instrument

- Under binary treatment/instrument, TSLS estimand is the LATE:

$$\tau = \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]} = \frac{\text{ITT}_Y}{\text{ITT}_D} = \tau_{\text{LATE}}$$

  - $\text{cov}(Y, Z)/\text{cov}(D, Z) = \beta_Y/\beta_D$ ($\mathbb{V}(Z)$ cancels, regardless of Z). For binary Z, OLS slope $\beta_{\cdot} = \mathbb{E}[\cdot|Z{=}1] - \mathbb{E}[\cdot|Z{=}0]$ (W4), giving $\text{ITT}_Y/\text{ITT}_D$.
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- And that the TSLS estimator is the Wald estimator:

$$\hat{\tau}_{\text{2SLS}} = \frac{\widehat{\text{cov}}(Y_i, Z_i)}{\widehat{\text{cov}}(D_i, Z_i)} = \frac{\overline{Y}_1 - \overline{Y}_0}{\overline{D}_1 - \overline{D}_0} = \frac{\widehat{\text{ITT}}_Y}{\widehat{\text{ITT}}_D} = \hat{\tau}_{\text{iv}}$$

$\rightsquigarrow$ Constant effects model is not required for TSLS in this setting.

TSLS with Covariates

- Adding covariates linearly:

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- Why not Lin (W4) fully-interacted models ($\tau + \gamma' \widetilde{\mathbf{X}}_i$ varies) in IV?
  1. Each $D_i \widetilde{X}_{ij}$ is endogenous and requires its own instrument, e.g., $Z_i \widetilde{X}_{ij}$.
  2. Strong prediction of D by Z does not imply strong prediction of $D_i \widetilde{X}_{ij}$ by $Z_i \widetilde{X}_{ij}$.
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- $\rightsquigarrow$ **PLIV** (§2) inherits constant τ (ML-flexible $g(\mathbf{X})$, $m(\mathbf{X})$ only). **DRIV** (§2) relaxes to $\tau(\mathbf{X})$.

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- The bias of the Wald estimator:

$$\widehat{\tau}_{\text{iv}} - \tau = \frac{\widehat{\text{cov}}(\tau D_i + \varepsilon_i, Z_i)}{\widehat{\text{cov}}(D_i, Z_i)} - \tau = \frac{\frac{1}{n}\sum_{i=1}^n \varepsilon_i Z_i}{\frac{1}{n}\sum_{i=1}^n \eta_i Z_i} \xrightarrow{d} \underbrace{\frac{\text{cov}(\varepsilon_i, \eta_i)}{\mathbb{V}[\varepsilon_i]}}_{\text{bias}} + \underbrace{W_i}_{\text{Cauchy r.v.}}$$

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- **Relevant IV**: noise divided by signal $\longrightarrow 0$; **Irrelevant IV**: noise divided by noise does not.

What to Do About Weak Instruments?

- **Detect**: first-stage F-test on excluded instruments ($H_0 : \gamma = 0$).
  - $F \geq 10 \Rightarrow$ bias is small (Stock & Yogo 2005); still the dominant convention in empirical IV.
  - $F \geq 104.7$ for correct CI coverage in the *worst-case* DGP (Lee et al. 2022).
    - Lee et al. show the standard 1.96 gives true 95% coverage *only* at $F \geq 104.7$; below that, actual coverage drops monotonically. At the conventional $F = 10$, worst-case actual coverage is around **85%**.

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- **Weak-IV robust inference**: Anderson-Rubin (1949) test (simplified setting, binary Z, D).
  - **Trick**: rewrite $H_0 : \tau = \tau_0$ as $H_0 : \text{ITT}_Y - \tau_0\, \text{ITT}_D = 0$: **linearises the null**, no weak-IV pathology.
  - Under the null, asymptotically:

$$g(\tau_0) = \widehat{\text{ITT}}_Y - \tau_0\, \widehat{\text{ITT}}_D \sim N(0, \Omega(\tau_0))$$
$$\Omega(\tau_0) = \mathbb{V}[\widehat{\text{ITT}}_Y] + \tau_0^2\, \mathbb{V}[\widehat{\text{ITT}}_D] - 2\tau_0 \operatorname{cov}(\widehat{\text{ITT}}_Y, \widehat{\text{ITT}}_D)$$

  - AR statistic: $g(\tau_0)^2/\Omega(\tau_0) \sim \chi_1^2$ regardless of first-stage strength; CI by analytical inversion.

```r
> library(ivmodel)
> m <- ivmodel(Y = y, D = D, Z = Z)
> m$AR    # AR test (F-stat), p-value, df, and 95% CI by inversion
```

  - Increasingly reported alongside the F-stat; handles weak-IV uncertainty correctly when first stage is borderline.

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  - Monotonicity: $D_i(1) \geq D_i(0)$ (instrument only increases treatment)
  - Exclusion restriction: $Y_i(1,d) = Y_i(0,d)$ for all $d = 0, 1, \ldots, K-1$

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- Can't identify the proportion of all compliance types here.
- Example: $K = 3 \rightsquigarrow 9$ principal strata
  - Affected: $(D_i(0), D_i(1)) \in \{(0,1), (0,2), (1,2)\}$
  - Unaffected: $(D_i(0), D_i(1)) \in \{(0,0), (1,1), (2,2)\}$
  - Negatively affected: $(D_i(0), D_i(1)) \in \{(1,0), (2,0), (2,1)\}$
  - Last ruled out by monotonicity.
  - 5 unknowns and 4 knowns under monotonicity.

TSLS with Multivalued Treatments

- Let $C_i = jk$ be an indicator for compliance type $D_i(1) = j$ and $D_i(0) = k$.
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- We can show that the 2SLS estimator converges to:

$$\widehat{\tau}_{2SLS} \xrightarrow{p} \sum_{k=0}^{K-1} \sum_{j=k+1}^{K-1} \omega_{jk} \mathbb{E}\left(\frac{Y_i(1) - Y_i(0)}{j-k} \,\middle|\, C_i = jk \right)$$

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- Intuition: a weighted average of effects per dose for each affected type.
  - Weights are proportional to size of the strata and how big the effect of the instrument is for that strata.
  - If instrument can only increase by 1 dose, then simplifies to weighted average of principal strata effects.

2/ Modern IV via Machine Learning

Cf. Chernozhukov et al. (2018), *Double/Debiased Machine Learning*, Econometrics Journal

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- Both share the engine **Double/Debiased ML** (Chernozhukov et al. 2018): ML for nuisances, Neyman-orthogonal moment, cross-fitting.

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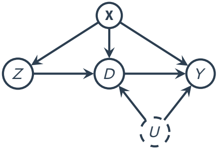

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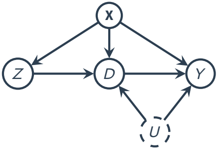

- **Three nuisance functions** to estimate:

$$\ell_0(\mathbf{X}) = \mathbb{E}[Y \mid \mathbf{X}], \qquad r_0(\mathbf{X}) = \mathbb{E}[D \mid \mathbf{X}], \qquad m_0(\mathbf{X}) = \mathbb{E}[Z \mid \mathbf{X}]$$

  - When $\mathbf{X} \not\to Z$ (i.e., $Z \perp \mathbf{X}$), m_0 collapses to the constant $\mathbb{E}[Z]$ and ML has nothing to do; it becomes a real function of $\mathbf{X}$ only when $\mathbf{X}$ predicts Z.

ML Nuisance Functions and High-Dim Confounding

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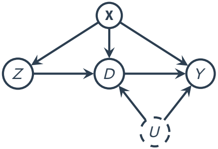

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- **Why naive plug-in fails**: ML regularization injects bias into $\widehat{\eta}$ that contaminates $\widehat{\tau}$. DML fixes this with two ideas:
  - **Neyman orthogonality**: moment is locally insensitive to nuisance errors at the truth.
  - **Cross-fitting**: fit $\widehat{\eta}$ on out-of-fold data, evaluate on held-out fold.

PLIV via DML: Algorithm and Conditions

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  1. Partition into folds $\{I_k\}_{k=1}^K$. For each k, fit $\widehat{\ell}_{[k]}, \widehat{r}_{[k]}, \widehat{m}_{[k]}$ on data *outside* I_k.
  2. For $i \in I_k$: $\widetilde{Y}_i = Y_i - \widehat{\ell}_{[k]}(\mathbf{X}_i)$, $\widetilde{D}_i = D_i - \widehat{r}_{[k]}(\mathbf{X}_i)$, $\widetilde{Z}_i = Z_i - \widehat{m}_{[k]}(\mathbf{X}_i)$.
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- **Validity conditions** (CHS 2018, Thm 13.1.1): strong IV after partialling out, plus ML estimates converging at $n^{1/4}$ rate.
- Software: `DoubleML` (R). Applied PLIV demo on AJR in §3.

From τ to $\tau(\mathbf{X})$: DRIV

- **DRIV** (Doubly Robust IV; Syrgkanis et al. 2019, *NeurIPS*): estimate the *function* $\tau(\mathbf{X})$ pointwise via doubly-robust orthogonal IV moment.
  - "How does the effect vary across covariates?"
  - "Doubly-robust" = the estimator stays consistent if either of two nuisance models is correctly specified.
  - **Implementation**: the NeurIPS authors released only a Python package (`EconML.DRIV`). R users call it from R via `reticulate`.

```r
# DRIV in R via reticulate (loads EconML Python under the hood)
# One-time setup:  reticulate::py_install("econml")
library(reticulate)
econml_dr <- import("econml.iv.dr")
sk_ens    <- import("sklearn.ensemble")

est <- econml_dr$DRIV(
  model_y_xw = sk_ens$RandomForestRegressor(),     # E[Y | X]   nuisance for outcome
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  cv = 5L                                          # K-fold cross-fitting
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est$fit(Y, T = D, Z = Z, X = X)         # cross-fitted nuisances + orthogonal moment
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3/ Estimating IV: AJR Walkthrough + Modern IV

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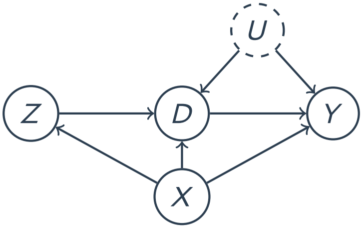
- DAG (recap from §2):



Property Rights & Economic Development

Created using DALL·E 3 motivated by Ratna Sagar Shrestha

Recognize the person on the right?

- Q: Do property rights (i.e., institutions) promote economic development?
    - Famous paper on this: Acemoglu, Johnson, and Robinson (2001) AER
    - Relationship between strength of property rights in a country and GDP.

The AJR Data

Name	Description
shortnam	three-letter country code
africa	indicator for if the country is in Africa
asia	indicator for if country is in Asia
logem4	log mortality rates faced by European settlers (IV)
avexpr	strength of property rights (protection against expropriation)
logpgp95	log GDP per capita

```
> ajr <- read_csv("https://bit.ly/3RUJDWK"); ajr

# A tibble: 163 × 15
   shortnam africa lat_abst malfal94 avexpr logpgp95 logem4  asia yellow baseco leb95
   <chr>     <dbl>    <dbl>    <dbl>  <dbl>    <dbl>  <dbl> <dbl>  <dbl>  <dbl> <dbl>
 1 AFG           0   0.367   0.00372 NA        NA      4.54     1      0     NA  NA
 2 AGO           1   0.137   0.950     5.36     7.77   5.63     0      1      1  46.5
 3 ARE           0   0.267   0.0123    7.18     9.80  NA        1      0     NA  NA
 4 ARG           0   0.378   0         6.39     9.13   4.23     0      0      1  72.9
 5 ARM           0   0.444   0        NA        7.68  NA        1      0     NA  NA
 6 AUS           0   0.300   0         9.32     9.90   2.15     0      1      1  78.2
 7 AUT           0   0.524   0         9.73     9.97  NA        0      0     NA  NA
 8 AZE           0   0.448   0        NA        7.31  NA        1      0     NA  NA
 9 BDI           1   0.0367  0.950    NA        6.57   5.63     0      1     NA  NA
10 BEL           0   0.561   0         9.68     9.99  NA        0      0     NA  NA
# i 153 more rows
# i 4 more variables: imr95 <dbl>, meantemp <dbl>, lt100km <dbl>, latabs <dbl>
# i Use `print(n = ...)` to see more rows
```

In R: Example Code Using ajr Data

```r
# Center (i.e., demeaning) the variables
> ajr <- ajr |>
    mutate(
      D_cnt = avexpr - mean(avexpr, na.rm = TRUE),
      Y_cnt = logpgp95 - mean(logpgp95, na.rm = TRUE),
      Z_cnt = logem4 - mean(logem4, na.rm = TRUE)
    ) |>
    na.omit()

# Compute the ITTs on D and on Y:
> ITT_D <- cov(ajr$Z_cnt, ajr$D_cnt)
> ITT_Y <- cov(ajr$Z_cnt, ajr$Y_cnt)
> wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
[1] 0.9242

# Same as reg Y~Z / reg D~Z:
> ITT_Y <- coef(lm(Y_cnt ~ Z_cnt, data = ajr))[[2]]
> ITT_D <- coef(lm(D_cnt ~ Z_cnt, data = ajr))[[2]]
> wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
[1] 0.9242
```

- The Wald estimate from manual calculation: 0.9242

In R: Example Code Using ajr Data

```
# Compare with ivreg
> ivreg_result <- AER::ivreg(Y_cnt ~ D_cnt | Z_cnt, data = ajr); summary(ivreg_result)

Call:
AER::ivreg(formula = Y_cnt ~ D_cnt | Z_cnt, data = ajr)

Residuals:
     Min       1Q   Median       3Q      Max
-2.40175 -0.54950  0.01792  0.68944  1.67361

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.031e-16  1.231e-01   0.000        1
D_cnt       9.242e-01  1.547e-01   5.974 1.59e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9458 on 57 degrees of freedom
Multiple R-Squared: 0.1107,        Adjusted R-squared: 0.09506
Wald test: 35.68 on 1 and 57 DF,  p-value: 1.589e-07

> round(coef(ivreg_result), digits = 4)
(Intercept)       D_cnt
     0.0000      0.9242
```

- The 2sls estimate from `AER::ivreg()`: 0.9242
  - Caveat: compute robust SE separately! (don't use SE from 2nd stage)

In R: Example Code Using ajr Data

```
# Estimate robust SE with estimatr::iv_robust()
> iv_rob <- estimatr::iv_robust(Y_cnt ~ D_cnt | Z_cnt, data = ajr, se_type = "HC2"); iv_rob
                Estimate Std. Error      t value     Pr(>|t|)   CI Lower CI Upper DF
(Intercept) 4.148820e-16  0.1234908 3.359619e-15 1.000000e+00 -0.2472860 0.247286 57
D_cnt       9.242173e-01  0.1791511 5.158871e+00 3.262823e-06  0.5654735 1.282961 57
```

- Robust SE with HC2 using `estimatr::iv_robust()`: 0.1791

```
> iv_rob_cov <- estimatr::iv_robust(Y_cnt ~ D_cnt + lat_abst + asia + africa |
                                     Z_cnt  + lat_abst + asia + africa, data = ajr,
                                   se_type = "HC2")
> var_labels <- c(
   "D_cnt" = "Avg. Expropriation Risk (D_cnt)", "lat_abst" = "Abs. Value of Latitude",
   "asia" = "Asian country", "africa" = "African country", "Z_cnt" = "log Mortality Rate (D_cnt)"
 )
```

In R: Example Code Using ajr Data

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# Estimate robust SE with estimatr::iv_robust()
> iv_rob <- estimatr::iv_robust(Y_cnt ~ D_cnt | Z_cnt, data = ajr, se_type = "HC2"); iv_rob
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 )
```

- When adjusting for covariates in TSLS, include them both in the 1st and 2nd stage.

IV Regression Table with `modelsummary`

```r
> modelsummary::modelsummary(
  models = list(ivreg_result, iv_robust, iv_rob_cov),
  coef_map = var_labels, gof_map = c("nobs", "r.squared", "adj.r.squared"), stars = T,
  notes = "Note: See appendix for other model statistics.", output = "modelsummary_tab.tex")
```

	(1)	(2)	(3)
Avg. Expropriation Risk (D_cnt)	0.924***	0.924***	1.015**
	(0.155)	(0.179)	(0.351)
Abs. Value of Latitude			-1.596
			(1.529)
Asian country			-1.048*
			(0.425)
African country			-0.390
			(0.342)
Num.Obs.	59	59	59
R2	0.111	0.111	0.045
R2 Adj.	0.095	0.095	-0.026

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: See appendix for other model statistics.

Modern IV in R: DoubleML for PLIV

- Same partialling-out as Modern IV section, in ∼10 R lines (cross-fitting built in):

```
> library(DoubleML); library(mlr3); library(mlr3learners)
> # Build DoubleMLData (y = outcome, d = treatment, z = instrument, x = covariates)
> data <- DoubleMLData$new(df, y_col = "Y", d_cols = "D",
+                              z_cols = "Z", x_cols = X_names)
> # ML learners for the nuisance functions
> ml_l <- lrn("regr.ranger")   # E[Y | X]
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> # PLIV with 5-fold cross-fitting  (Neyman orthogonality + cross-fitting = DML)
> pliv <- DoubleMLPLIV$new(data, ml_l, ml_m, ml_r, n_folds = 5)
> pliv$fit(); pliv$summary()
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- Also: $\widehat{\tau}$ sensitive to learner / $\mathbf{X}$ choice (Ahrens et al. 2025 §5–6, by DML 2018 authors). Remedies proposed; revisit in **W15**.

Modern IV Demo: A Digital Advertising Experiment

- Online book retailer A/B-tests two ad formats ($n \approx 1.4$M user-impressions, 76 ad creatives).
  - Z: random assignment to interactive (Blind Lead-Ins, BLI) vs. static banner.
  - D: BLI *completion* (treated-side completion rate ≈ 0.32; **one-sided noncompliance**: $D = 0$ for all $Z = 0$).
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- Why this is a useful DML-IV testbed:
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- **Plan**: (i) classical pooled 2SLS LATE (`fixest`) as baseline; (ii) Causal Forest IV $\rightsquigarrow \widehat{\tau}(\mathbf{X})$ via `grf::instrumental_forest` (R-native HTE-IV from §2; same target as DRIV, distinct moment construction).

Pooled 2SLS LATE: Classical Baseline

- 2SLS LATE on 30-day conversion (date FE, user covariates, cluster-robust SE at the user level):

```
> library(fixest)
> late <- feols(lngt_conv ~ ctrl_vars | date | tot ~ treatment,
+               data = d, cluster = ~ memno)
> summary(late)
TSLS estimation, Dep. Var.: lngt_conv
Observations: 1,408,121 / Fixed-effects: date (80)
Standard-errors: Clustered (memno)
        Estimate Std. Error t value Pr(>|t|)
fit_tot 0.002326   0.001210   1.922   0.0548 .
... (control coefficients omitted)
First-stage F (Kleibergen-Paap): ~1.5e4   # very strong instrument
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- **LATE**: BLI completion raises 30-day conversion by ≈ 0.23 **percentage points** among compliers (`fit_tot` = 0.00233, $p \approx 0.055$).

From $\widehat{\tau}$ to $\widehat{\tau}(\mathbf{X})$: Causal Forest IV in R

- R-native HTE-IV (distinct from DRIV; §2): `grf::instrumental_forest` (Athey–Tibshirani–Wager 2019); **X** = 4 user + 4 creative features.

```r
> library(grf)
> ivf <- instrumental_forest(
+   X = X_matrix, Y = Y, W = D, Z = Z,
+   num.trees = 4000, min.node.size = 100, honesty = TRUE)
> average_treatment_effect(ivf)              # forest-weighted ATE on BLI sample
> predict(ivf, X_new)$predictions            # tau-hat(X) at any covariate point
> best_linear_projection(ivf, A = X_matrix)  # which features drive HTE; valid SE
> test_calibration(ivf)                      # is HTE detectable above noise?
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- Same target $\widehat{\tau}(\mathbf{X})$ as DRIV (§2), built from forest-weighted local 2SLS rather than the orthogonal-score moment.

What Causal Forest IV Reveals on the BLI Data

- **Pooled 2SLS LATE**: BLI completion raises 30-day conversion by ≈ 0.23 percentage points among compliers (just-id binary IV, one-sided noncompliance).

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- **Subgroup CATE**: engaged users (high recent visit-days) absorb ≈ 0.39 pp; disengaged users ≈ 0.05 pp. *Where flexible HTE-IV adds value beyond the pooled LATE: a near-8$\times$ gap that the single-number 2SLS estimate hides.*

4/ Notable IV Designs in Recent Empirical Work

Source: Causal Inference for the Brave and True By Matheus Facure Alves

Treatment Assignments as IVs

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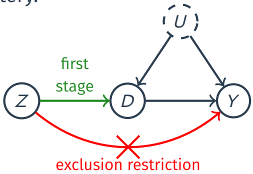
- Context: 9-month RCT in Ctrip's Shanghai call-centre; 249 volunteers entered the lottery.



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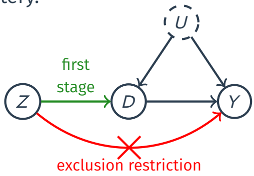

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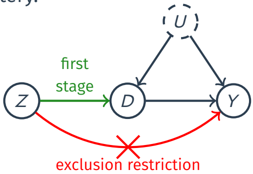

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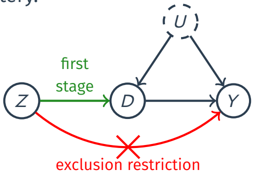

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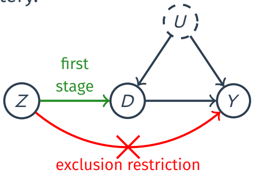

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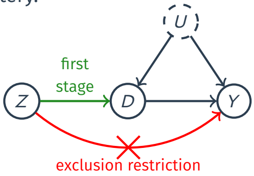

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- IV logic:
  1. Randomization: Balance tests on 18 baseline characteristics show no joint differences; p=0.466.
  2. Relevance: 80-90% of winners comply (1st-stage F $\approx$ 23)
  3. Exclusion: tasks, pay, IT identical – only location changes.
  4. Monotonicity: winners may or may not take WFH, losers cannot $\rightsquigarrow$ $D_i(1) \geq D_i(0)$ trivially (one-sided non-compliance).

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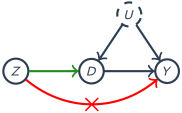

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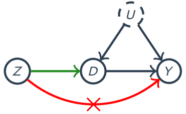

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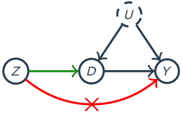

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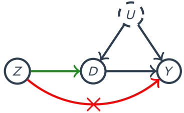

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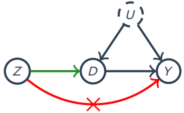

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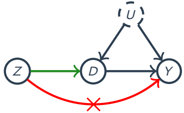

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- IV logic:
  1. Randomization: 90% of cell-towers built before 2010 and shows only weak correlations with 2010–15 population growth and store count, included in $\mathbf{X}_i$.
  2. Relevance: Each cell tower raises the log-odds of adoption by 0.0022 (z=3.7).
  3. Exclusion: controlled for ZIP income, store distance, and competitor presence to block direct demand channels.
  4. Monotonicity: Better signal only increases the prob. of downloading the app (no defiers).

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Peers' Environments as IVs

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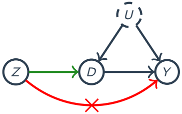

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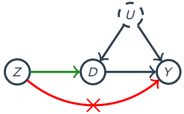

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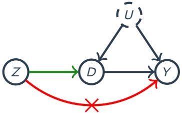

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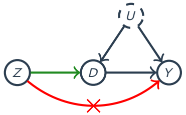

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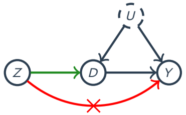

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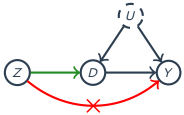

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- IV logic:
  1. Randomization: Use only friend pairs in different cities whose weather paths are uncorrelated; ego's own weather + date FE included.
  2. Relevance: first-stage F = 216–430 well above Stock-Yogo cutoff.
  3. Exclusion: Weather in friend's city is uncorrelated with ego's weather by construction.
  4. Monotonicity: bad weather never makes the friend run more.

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Onto the presentations & discussions!

Contact Information:

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https://j1yoo.github.io/

Appendix

General 2SLS

- Linear model for each i:

$$Y_i = \mathbf{X}_i'\beta + \varepsilon_i$$

    - $\mathbf{X}_i$ is $k \times 1$ and now includes D_i and any pretreatment covariates.
    - Parts of $\mathbf{X}_i$ are endogenous so that $\mathbb{E}[\varepsilon_i \mid \mathbf{X}_i] \neq 0$

- Instruments $\mathbf{Z}_i$ that is $\ell \times 1$ vector such that $\mathbb{E}[\varepsilon_i \mid \mathbf{Z}_i] = 0$.
    - $\mathbf{Z}_i$ might include exogenous/pretreatment variables from $\mathbf{X}_i$ as well.
    - Rank condition: $\mathbb{E}[\mathbf{Z}_i\mathbf{Z}_i']$ and $\mathbb{E}[\mathbf{X}_i\mathbf{Z}_i']$ have full rank.

- Identification:
    - $k = \ell$: just-identified.
    - $k < \ell$: over-identified (can test the exclusion restriction, kinda)
    - $k > \ell$: unidentified (fails rank condition)

Nasty Matrix Algebra

- Projection matrix projects values of $\mathbf{X}_i$ onto $\mathbf{Z}_i$:

$$\begin{aligned}
\mathbf{\Pi} &= (\mathbb{E}[\mathbf{Z}_i\mathbf{Z}_i'])^{-1}\,\mathbb{E}[\mathbf{Z}_i\mathbf{X}_i'] \quad \text{(projection matrix, i.e., pi/}\mathbf{\Pi}\text{)}\\
\tilde{\mathbf{X}}_i &= \mathbf{\Pi}'\mathbf{Z}_i \quad \text{(projected values)}
\end{aligned}$$

- To derive the 2SLS estimator, take the fitted values, $\mathbf{\Pi}'\mathbf{Z}_i$ and multiply both sides of the outcome equation by them:

$$\begin{aligned}
Y_i &= \mathbf{X}_i'\beta + \varepsilon_i\\
\mathbf{\Pi}'\mathbf{Z}_i Y_i &= \mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i'\beta + \mathbf{\Pi}'\mathbf{Z}_i\varepsilon_i\\
\mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i Y_i] &= \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i']\beta + \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\varepsilon_i]\\
\mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i Y_i] &= \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i']\beta + \mathbf{\Pi}'\mathbb{E}[\mathbf{Z}_i\varepsilon_i]\\
\mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i Y_i] &= \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i']\beta\\
\mathbb{E}[\tilde{\mathbf{X}}_i Y_i] &= \mathbb{E}[\tilde{\mathbf{X}}_i\mathbf{X}_i']\beta\\
\beta &= \left(\mathbb{E}[\tilde{\mathbf{X}}_i\mathbf{X}_i']\right)^{-1}\mathbb{E}[\tilde{\mathbf{X}}_i Y_i]
\end{aligned}$$

How to Estimate the Parameters

- Collect $\mathbf{X}_i$ into an $n \times k$ matrix $\mathbb{X} = (\mathbf{X}_1', \ldots, \mathbf{X}_n')$
- Collect $\mathbf{Z}_i$ into an $n \times \ell$ matrix $\mathbb{Z} = (\mathbf{Z}_1', \ldots, \mathbf{Z}_n')$

- In-sample projection matrix produces fitted values:

$$\widehat{\mathbb{X}} = \mathbb{Z}(\mathbb{Z}'\mathbb{Z})^{-1}\mathbb{Z}'\mathbb{X}$$

  - Fitted values of the regression of $\mathbb{X}$ on $\mathbb{Z}$.
  - Matrix party trick: $\mathbb{X}'\mathbb{Z}/n = (1/n)\sum_i^n \mathbf{X}_i\mathbf{Z}_i' \xrightarrow{p} \mathbb{E}[\mathbf{X}_i\mathbf{Z}_i']$.

- Take the population formula for the parameters:

$$\beta = \left(\mathbb{E}[\tilde{\mathbf{X}}_i\mathbf{X}_i']\right)^{-1}\mathbb{E}[\tilde{\mathbf{X}}_i Y_i]$$

- And plug in the sample values (the n cancels out):

$$\widehat{\beta}_{2SLS} = (\widehat{\mathbb{X}}'\mathbb{X})^{-1}\widehat{\mathbb{X}}'\mathbf{y} \xrightarrow{p} \beta$$

- This is how `R`/`Stata` estimate the 2SLS parameters.

Asymptotic Variance for 2SLS

- We can write the centered, normalized TSLS estimator as:

$$\sqrt{n}\left(\widehat{\beta}_{2\text{SLS}} - \beta\right) = \underbrace{\left(n^{-1}\sum_i \widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i'\right)^{-1}}_{\xrightarrow{p} (\mathbb{E}[\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i'])^{-1}} \underbrace{\left(n^{-1/2}\sum_i \widehat{\mathbf{X}}_i\varepsilon_i\right)}_{\xrightarrow{d} N\left(0, \mathbb{E}[\widehat{\mathbf{X}}_i'\varepsilon_i'\varepsilon_i\widehat{\mathbf{X}}_i]\right)}$$

- Thus, $\sqrt{n}\left(\widehat{\beta}_{2\text{SLS}} - \beta\right)$ has asymptotic variance:

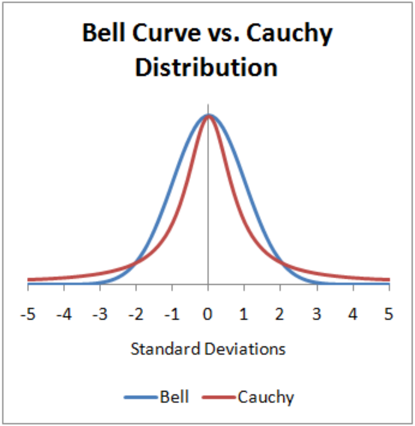
$$\left(\mathbb{E}[\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i']\right)^{-1} \mathbb{E}[\widehat{\mathbf{X}}_i'\varepsilon_i'\varepsilon_i\widehat{\mathbf{X}}_i] \left(\mathbb{E}[\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i']\right)^{-1}$$

- **Robust 2SLS variance estimator** with residuals $\widehat{u}_i = Y_i - \mathbf{X}_i'\widehat{\beta}$:

$$\widehat{\text{Var}}\left(\widehat{\beta}_{2\text{SLS}}\right) = (\widehat{\mathbb{X}}'\widehat{\mathbb{X}})^{-1}\left(\sum_i \widehat{u}_i^2\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i'\right)(\widehat{\mathbb{X}}'\widehat{\mathbb{X}})^{-1}$$

- HC2, clustering, and autocorrelation versions exist.

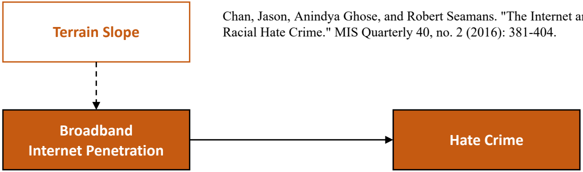
Cauchy vs. Normal Distribution



Source: https://stats.stackexchange.com/questions/36027/why-does-the-cauchy-distribution-have-no-mean

Regional Characteristics as IVs

- Chan, Ghose, Seamans, MISQ, 2016:

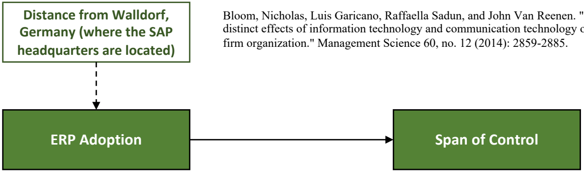


Chan, Jason, Anindya Ghose, and Robert Seamans. "The Internet and Racial Hate Crime." MIS Quarterly 40, no. 2 (2016): 381-404.

- Observational study using *Hate Crime Statistics* from FBI.
  - RQ: Does the spread of Internet increase racial hate crime?
- Issue? Confounding
- IV? terrain slope/steepness (i.e., how many hills in a given region?)

Geographical Proximity-Based IVs

- RQ: Does the adoption of an ERP system impact plant manager autonomy (i.e., span of control)?
  - Span of control: the number of employees managed by supervisors or managers in an organization (high SoC = centralized).

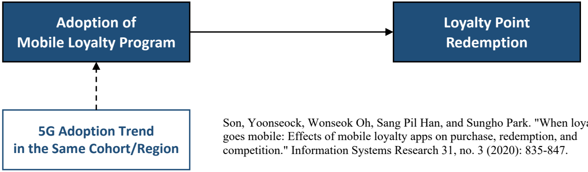


Bloom, Nicholas, Luis Garicano, Raffaella Sadun, and John Van Reenen. "The distinct effects of information technology and communication technology on firm organization." Management Science 60, no. 12 (2014): 2859-2885.

  - Observational study: the CEP management and organization survey and the Harte-Hanks ICT panel (Bloom, Garicano, Sadun, Van Reenen, MgmtSci, 2014).
  - IV: Distance from the ERP market leader (i.e., SAP) w/ 25%+ market share ⇝ likely more established connections with SAP (German firms vs. French, England firms).

Macro/Cohort Trends as IVs

- RQ: How does loyalty app adoption affect customer point redemption behavior?

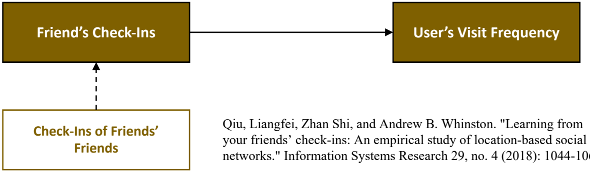


Son, Yoonseock, Wonseok Oh, Sang Pil Han, and Sungho Park. "When loyalty goes mobile: Effects of mobile loyalty apps on purchase, redemption, and competition." Information Systems Research 31, no. 3 (2020): 835-847.

  - Observational study: data on loyalty app adoption status, loyalty point redemption patterns, and purchase behaviors in multivendor loyalty program (MVLP) context (Son, Oh, Han, Park, ISR, 2020).
  - IV: 5G adoption rate in the same cohort (e.g., age group).

Peers' Environments as IVs

- Qiu, Shi, Whinston, ISR, 2018:



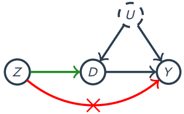
Qiu, Liangfei, Zhan Shi, and Andrew B. Whinston. "Learning from your friends' check-ins: An empirical study of location-based social networks." Information Systems Research 29, no. 4 (2018): 1044-1061.

  - Observational study using data on restaurant check-in information and the users' social network ties from a major SNS in China.
    - RQ: Is there observational learning/herding effect for restaurant discovery?
  - Issue? Homophily.
  - IV: Check-in activities of friend's friends.

Bartik (Shift-Share) IVs

Barron et al. (*Marketing Sci.* 2020)[4]

- Context: Panel of 43,000 ZIP codes (100 largest U.S. metro areas), monthly 2011-2016. Airbnb listings scraped; Zillow rent & price indices matched at ZIP-month.



Instrument Z: Google-Trends "Airbnb" (global shock) × 2010 ZIP "touristiness" (# food/lodging firms)

Treatment D: $\ln(1 + \text{Airbnb listings})$ in ZIP-month.

Outcome Y: Zillow rent index, house price index , price-to-rent.

- Identification issue: Hot ZIPs attract both Airbnb supply and rising housing costs $\rightsquigarrow$ upward bias.
- IV logic:
  1. Conditional exogeneity: 2010 touristiness fixed before Airbnb's national growth; Google-Trends shock is global, not driven by ZIP-specific factors. **(Hardest assumption for Bartik IVs; defended in detail next slide.)**
  2. Relevance: first-stage F = 650–820 well above Stock-Yogo cutoff.
  3. Exclusion: global search shocks unrelated to local housing; touristiness fixed in 2010. City×month FE + placebo ZIPs (no listings) show no direct price effect.
  4. Monotonicity: more global Airbnb awareness cannot lower listings in touristy ZIPs.

[4] Barron, K., Kung, E., & Proserpio, D. "The Effect of Home-Sharing on House Prices and Rents" Marketing Sci. 2020. 40(2): 283–304.

Bartik (Shift–Share) IVs

Barron et al. (*Marketing Sci.* 2020)

- IV logic 1 (continued): defending **conditional exogeneity**, the hardest assumption for Bartik IVs:
    - Parallel pre-trends: prior to 2012, rents and prices evolve identically across touristiness quartiles (Fig. 5) $\rightsquigarrow$ no pre-existing divergence.
    - Touristiness $\times$ time test: adding a ZIP-specific trend term ($h_{i,2010} \times t$) yields an insignificant coefficient; IV effect unchanged.
    - Rich time-varying controls: results robust after adding ZIP income, population, hotel rooms, TripAdvisor reviews, and airport arrivals $\rightsquigarrow$ IV not picking up gentrification or tourism shocks.